Restricted contraction for preconditioned plug-and-play: exact quasi-Newton spectra and compressed-sensing recovery with learned denoisers

Viktor Vigren*

Draft, November 10, 2026

Abstract

Plug-and-play (PnP) reconstruction alternates a data-consistency step with a learned denoiser. For preconditioned PnP in compressed-sensing MRI, the available convergence theory requires $(1+\epsilon)\,\rho(I-\alpha PA^*A) < 1$. We show that this condition cannot hold when A has a nontrivial null space, that it is not sufficient as stated when PA^*A is not normal, and that the constant in the companion stability bound for the zero-memory self-scaling symmetric rank-one (ZMSHR1) preconditioner is pessimistic by a factor $\approx 1/(2\delta_{\mathrm{sg}}) = 5\cdot 10^7$ for the published parameters, because the safeguard behind it can never activate. These facts follow from an exact formula for the ZMSHR1 matrix: its eigenvalues are $\rho_k(1\pm\sin\vartheta_k)$, with ρ_k the Barzilai–Borwein step and ϑ_k the secant angle. We then replace global contraction by *restricted* contraction. If the denoiser's secant acts on the error as a nonexpansive map supported on a d-dimensional tangent space, up to a leakage η, then PnP approaches the truth at a linear rate whenever a pairwise restricted constant δ of the data step satisfies $\eta m(1-\delta+m) < 1-\delta$, where m bounds the data step. The error bound is explicit in noise, model mismatch and leakage, and holds for undersampled operators and dynamic preconditioners, without a fixed-point assumption. For random block sampling (non-Cartesian, multi-coil), a matrix Chernoff bound yields the restricted condition with high probability once the number of shots exceeds a multiple of $(\Phi_0/\lambda_-)\log(d/p)$: the sample complexity is proportional to the denoiser's tangent dimension, scaled by a block coherence. Finally, preconditioning changes the PnP *estimator*, not only the convergence path. Locally, the noise part of the PnP limit is the best linear unbiased estimator on the tangent space if and only if P maps no data-visible direction orthogonal to the tangent space into it. The model mismatch inside the tangent space, however, is amplified by $(\alpha\Pi_0 PH\Pi_0)^{-1}$, which is up to the restricted condition number for $P = I$. Density-compensation-type preconditioners fail on both counts. A tangent-restricted preconditioner attains both the optimal variance and the unamplified bias. Toy experiments in which every hypothesis can be computed illustrate each result. [CHECK: abstract length for the target journal]

1 Introduction

Let $A : \mathbb{R}^n \to \mathbb{R}^q$ be a linear measurement operator, $y = Ax^\natural + n$ the data, $f(x) = \frac{1}{2}\|Ax-y\|^2$ and $H = A^*A$. Plug-and-play (PnP) methods [43, 26] replace the proximal step of a splitting method by a denoiser D. The simplest instance, PnP-ISTA, and its preconditioned variant P^{2}nP [20] read

$$x_{k+1} = D\big(z_k\big), \qquad z_k = x_k - \alpha P_k \nabla f(x_k) = x_k - \alpha P_k A^*(Ax_k - y), \tag{1}$$

*[CHECK: affiliation, WASP, funding acknowledgement, co-authors]

with $P_k \succ 0$ fixed or updated from iterate differences. In MRI, A is a multi-coil, often non-Cartesian Fourier operator. Without density compensation, H is very ill-conditioned (the centre of k-space is sampled far more densely than the periphery), and a preconditioner can reduce the iteration count by an order of magnitude [20].

The convergence theory of [20] follows the template of [39]: the denoiser is $(1+\epsilon)$-Lipschitz, the data step $I - \alpha PH$ is a contraction, and the product contracts if $(1+\epsilon)\rho(I-\alpha PH) < 1$. For a dynamic quasi-Newton preconditioner (ZMSHR1, built on [34, 5, 6, 19]) a stability bound is given whose constant depends on the safeguard parameter as $1/\delta_{\mathrm{sg}}$. The authors note that their bound becomes very large for the parameters that work well, and that "further analysis could lead to tighter results" [20, Sec. III].

This paper supplies that analysis and goes further. Its starting point is that, in compressed sensing, contraction can only come from the *product* of denoiser and data step. The denoiser must remove exactly the directions that the data cannot see. Any argument that bounds the two factors separately is therefore vacuous when A has a null space. The right objects are the denoiser's (secant) Jacobian, which for trained denoisers is close to a projector onto a low-dimensional, signal-adaptive subspace [32, 25, 24], and the data step *restricted* to such subspaces.

Contributions.

1. **The existing conditions** (Section 2). The condition $(1+\epsilon)\rho(I-\alpha PH) < 1$ is unsatisfiable when $\operatorname{null} A \neq \{0\}$ (Observation 2.1). It is also not sufficient when PH is not normal: we give a two-dimensional counterexample with a 1-Lipschitz denoiser and divergent iterates (Proposition 2.2). The operator norm restores the statement.

2. **Exact ZMSHR1 spectrum** (Section 3). The ZMSHR1 matrix has eigenvalues $\rho(1-\sin\vartheta)$ (multiplicity $n-1$) and $\rho(1+\sin\vartheta)$ (Lemma 3.1). It follows that the safeguard of [20] is inert unless $\delta_{\mathrm{sg}} \geq \frac{1}{2}\sqrt{\theta_1/\theta_2}$ (Corollary 3.3), that $P_k \preceq \frac{1+\sqrt{1-\theta_1/\theta_2}}{\theta_1} I$ with no dependence on δ_{sg}, and that the stability constant improves by a factor $\approx 1/(2\delta_{\mathrm{sg}})$ (Theorem 3.4).

3. **Restricted contraction** (Section 4). Under a secant tangent-structure assumption with leakage η (Assumption 4.1), PnP with arbitrary (dynamic) preconditioners satisfies a coupled two-term recursion for the error to the *truth*. It converges linearly to an explicit ball when $\eta m(1-\delta+m) < 1-\delta$ (Theorem 4.5). Soft thresholding satisfies the assumption with $\eta = 0$ globally (Example 4.3). The exact local rate is $\rho\big(J_D(z_*)(I-\alpha PH)\big)$ (Proposition 4.7).

4. **Sampling theorems** (Section 5). For i.i.d. block sampling, the restricted constant is controlled with high probability once the number of shots is at least $\frac{3\Phi_0}{t^2\lambda_-}\log\frac{2d}{p}$. Here Φ_0 is a block coherence of the tangent space at the truth and λ_- the restricted expected curvature (Theorem 5.2). A uniform version over a family of tangent spaces (Theorem 5.5) addresses the "moving sparsifying basis" of a learned prior.

5. **Preconditioners change the estimator** (Section 6). In a local affine model of the denoiser, we give the PnP limit with preconditioner P in closed form (Proposition 6.1):

   - its noise part is the best linear unbiased estimator on the tangent space if and only if $\Pi_0 PA^*(I-Q_0) = 0$;
   - its in-tangent bias is the mismatch multiplied by $(\alpha\Pi_0 PH\Pi_0)^{-1}$.

   $P = I$ minimises the variance, but not the bias; a tangent-restricted preconditioner does both. We also show that ZMSHR1 is automatically well bounded on restricted secants, so capping it through θ_1 is unnecessary and slows it down (Proposition 6.4).

Related work.

- *Convergence of PnP* via (firmly) nonexpansive or contractive denoisers, or denoisers that are gradients or proximal maps: [39, 12, 35, 21, 15]. These give convergence to a fixed point, not recovery of the truth, and their contraction hypotheses fail for normalization-equivariant denoisers [18], whose Jacobian has the eigenvalue 1 on constants.

- *Recovery guarantees for PnP and projected-gradient schemes* with approximate projections [30, 41, 23, 28, 45] combine a restricted isometry on the secant set $\Sigma - \Sigma$ of a model set with a Lipschitz or restricted-Lipschitz property of the denoiser. When A is strongly undersampled, a bound on $\|(I - B)v\|$ over all secants v forces the denoiser to be within $O(\operatorname{rank} A/n)$ of a nonexpansive projection [CHECK: state and cite the precise form; our survey note derives it]. The tangent-space formulation used here avoids this. It is the denoiser analogue of the Riemannian analysis of low-rank recovery [44], and of the local linear convergence of forward–backward methods under partial smoothness [29], which in turn generalises the local analysis of iterative soft thresholding [11, 17].

- *Sampling theory* for structured and variable-density Fourier measurements [27, 2], for generative priors [9, 7, 1], and for unions of subspaces and manifolds [8, 3, 14]. The block coherence of Section 5 is the "variation" of [1], specialised to a tangent space. Sample complexity proportional to a *local* rank also appears for Gaussian mixture priors [38]. To our knowledge no sampling theorem for a *denoiser*-defined prior under Fourier sampling was available [CHECK: re-verify before submission].

- *Quasi-Newton.* The self-scaling in ZMSHR1 is the optimally conditioned choice of [34, 13]. The exact spectrum in Lemma 3.1 may be implicit there [CHECK: verify against Osborne–Sun 1999 before claiming novelty]. What is new is its use to sharpen the PnP analysis.

Notation.

- We work over a real Euclidean space $\mathbb{R}^n$. Complex images and complex data are identified with $\mathbb{R}^{2N}$ and $\mathbb{R}^{2q'}$, so A is real-linear and A^* is its real adjoint ($\operatorname{Re} A^H$ for complex A). All statements of Section 3 also hold verbatim over $\mathbb{C}$ with Hermitian inner products.

- $x^\natural$ is the truth and x_k the iterates. $\|\cdot\|$ is the Euclidean norm and the induced operator norm, and $\rho(\cdot)$ the spectral radius. Π_T is the orthogonal projector onto a subspace T, and $\lambda_{\min}(X|_T)$ the smallest eigenvalue of a symmetric X compressed to T.

- $L = \|H\|$. Noise is $n \sim \mathcal{N}(0, \sigma^2 I)$ in real coordinates.

2 Setting and the limits of global contraction

We write the error to the truth as $e_k = x_k - x^\natural$, the data step as $M_k = I - \alpha P_k H$, and the back-projected noise as $w_k = \alpha P_k A^* n$. Then

$$z_k - x^\natural = M_k e_k + w_k. \tag{2}$$

The results of [20]. We restate them in our notation (their Assumption 1, Theorems 1 and 2, Lemma 1):

- **Assumption.** D is $(1+\epsilon)$-Lipschitz on all of $\mathbb{C}^N$, with $\epsilon > 0$.
- **Theorem 1 (fixed P).** If $r_0 = (1+\epsilon)\rho(I - \alpha PH) < 1$, the iterates converge to a fixed point at rate at most r_0.
- **Theorem 2 (dynamic P_k).** Suppose $0 \prec P_* \preceq \lambda_* I$, $q = (1+\epsilon)\rho(I - \alpha P_* H) < 1$, $\|\nabla f(x_k)\| \le R$ and $x_* = D(x_* - \alpha P_* \nabla f(x_*))$. Then

$$\|x_{k+1} - x_*\| \le q^k \|x_1 - x_*\| + \frac{(1+\epsilon)\alpha(\delta_{\mathrm{sg}} + 1 + \delta_{\mathrm{sg}}\theta_1\lambda_*)}{\delta_{\mathrm{sg}}\theta_1(1-q)} R.$$

  This rests on the bound $P_k \preceq \frac{\delta_{\mathrm{sg}}+1}{\delta_{\mathrm{sg}}\theta_1} I$.

Observation 2.1 (vacuity under undersampling)**.** *If* $\operatorname{null} A \neq \{0\}$*, then* $\rho(I - \alpha PH) \ge 1$ *for every* $P \succ 0$ *and* $\alpha > 0$*, so* $r_0, q \ge 1 + \epsilon > 1$*. If* A *is injective, then* $r_0 < 1$ *requires* $\epsilon < \frac{\alpha\lambda_{\min}(PH)}{1 - \alpha\lambda_{\min}(PH)}$*. For an ill-conditioned multi-coil* H *this means an essentially nonexpansive denoiser.*

Proof. For $v \in \operatorname{null} A$, $(I - \alpha PH)v = v$. If A is injective, PH is similar to $P^{1/2} H P^{1/2} \succ 0$, so $\rho(I - \alpha PH) \ge 1 - \alpha\lambda_{\min}(PH)$. □

A normalization-equivariant denoiser [18], as used in [20], satisfies $J_D \mathbf{1} = \mathbf{1}$ and hence has Lipschitz constant at least 1. In the undersampled regime the theory therefore covers no denoiser at all. The obstruction is structural: the bound $\|T(x) - T(x')\| \le \operatorname{Lip}(D)\, \|I - \alpha PH\|\, \|x - x'\|$ treats the two factors as independent, while in compressed sensing contraction arises only because the denoiser suppresses the null-space directions on which $I - \alpha PH$ is the identity.

The proof of Theorem 1 in [20] bounds the data step by $\|I - \alpha PH\|$, and only the last line replaces it by ρ. The two agree when PH is normal, e.g. for polynomial preconditioners $P = p(H)$, which are the fixed preconditioners used there. In general they differ, and the statement with ρ is false:

Proposition 2.2 (counterexample for non-normal PH)**.** *Let* $n = 2$*,* $H = \left(\begin{smallmatrix} 1 & -2 \\ -2 & 5 \end{smallmatrix}\right) \succ 0$*,* $P = \left(\begin{smallmatrix} 1 & 0.99 \\ 0.99 & 1 \end{smallmatrix}\right) \succ 0$*,* $\alpha = 0.4$*,* $x^\natural \in \mathbb{R}^2$*,* $y = Ax^\natural$ *for any* A *with* $A^*A = H$*, and let* $D(z) = \Pi z + (I - \Pi)x^\natural$ *with* $\Pi = \mathsf{e}_1 \mathsf{e}_1^\top$*.*

- D *is* 1*-Lipschitz, so the assumption of Theorem 1 holds for every* $\epsilon > 0$*.*
- $\rho(I - \alpha PH) = 0.99608$*, so* $r_0 < 1$ *for every* $\epsilon < 0.0039$*.*
- *Nevertheless* $x_k - x^\natural = (1.392)^{k-1} \langle \mathsf{e}_1, x_1 - x^\natural \rangle\, \mathsf{e}_1$ *for* $k \ge 1$*, so the iterates diverge from every starting point with* $\langle \mathsf{e}_1, x_1 - x^\natural \rangle \neq 0$*, although* $x^\natural$ *is the unique fixed point.*

Proof. After one step, $x_k \in x^\natural + \operatorname{range}\Pi$, and $e_{k+1} = \Pi(I - \alpha PH)e_k = (1 - \alpha(PH)_{11})e_k$ with $(PH)_{11} = 1 - 0.99 \cdot 2 = -0.98$. The eigenvalues of PH are ≈ 2.0302 and ≈ 0.0098, both positive, which gives $\rho(I - \alpha PH)$. The fixed point is unique because $\Pi(PH)\Pi \neq 0$ on $\operatorname{range}\Pi$. □

The theorem holds as stated with $\|I - \alpha PH\|$ in place of ρ. Alternatively it holds with ρ if the Lipschitz assumption on D is made in the P^{-1}-norm, in which $I - \alpha PH$ is self-adjoint. Since $\|I - \alpha PH\| \le \sqrt{\kappa(P)}\, \rho(I - \alpha PH)$, the two versions differ by up to $\sqrt{\kappa(P)}$. Proposition 2.2 also shows something that recurs in Section 6: the relevant object is the data step *compressed to the range of the denoiser's Jacobian.* Here it is $\Pi(I - \alpha PH)\Pi$, and its eigenvalue can exceed 1 although $I - \alpha PH$ is a spectral contraction.

3 The exact spectrum of the ZMSHR1 preconditioner

Given $s = x_k - x_{k-1}$ and $h = \nabla f(x_k) - \nabla f(x_{k-1}) = Hs$, ZMSHR1 [20, Alg. 2] proceeds as follows.

1. **Mixing.** Choose the smallest $a \in [0,1]$ such that $v = as + (1-a)h$ satisfies $\theta_1 \le \langle s, v\rangle / \|s\|^2$ and $\|v\|^2/\langle s, v\rangle \le \theta_2$.
2. **Scaling.** Set $\tau = \frac{\|s\|^2}{\langle s,v\rangle} - \big(\frac{\|s\|^4}{\langle s,v\rangle^2} - \frac{\|s\|^2}{\|v\|^2}\big)^{1/2}$ and $u = s - \tau v$.
3. **Safeguard.** If $\langle u, v\rangle \le \delta_{\mathrm{sg}}\|u\|\|v\|$, set $u = 0$.
4. **Update.** Return $P = \tau I + \frac{uu^*}{\langle u,v\rangle}$.

The published parameters are $\delta_{\mathrm{sg}} = 10^{-8}$, $\theta_1 = 2 \cdot 10^{-6}$ and $\theta_2 = 200$.

Lemma 3.1 (spectrum)**.** *Let $s, v \in \mathbb{R}^n$ (or $\mathbb{C}^n$) with $\langle s, v\rangle > 0$, and define*

$$\rho = \frac{\|s\|^2}{\langle s, v\rangle}, \qquad \cos\vartheta = \frac{\langle s, v\rangle}{\|s\|\|v\|} \in (0, 1].$$

Let τ and u be as above. Then

1. $\tau = \rho(1 - \sin\vartheta) = \dfrac{\langle s, v\rangle}{\|v\|^2(1 + \sin\vartheta)}$;
2. $\langle u, v\rangle = \langle s, v\rangle \dfrac{\sin\vartheta}{1 + \sin\vartheta} \ge 0$ *and* $\cos\angle(u, v) = \dfrac{\cos\vartheta}{\sqrt{2(1 + \sin\vartheta)}}$ *(for $u \ne 0$);*
3. *if the safeguard does not fire, $P = \tau I + uu^*/\langle u, v\rangle$ satisfies the secant equation $Pv = s$, and its eigenvalues are $\rho(1 - \sin\vartheta)$ on $u^\perp$ and $\rho(1 + \sin\vartheta)$ on u.*

Proof. **Item 1.** Write $r_2 = \|s\|^2/\|v\|^2$. Then $\rho^2 - r_2 = \rho^2(1 - \cos^2\vartheta)$, so $\tau = \rho - \sqrt{\rho^2 - r_2} = \rho(1 - \sin\vartheta)$. The second form follows from $1 - \sin\vartheta = \cos^2\vartheta/(1 + \sin\vartheta)$ and $\rho\cos^2\vartheta = \langle s, v\rangle/\|v\|^2$.

Item 2. From item 1, $\tau\|v\|^2 = \langle s, v\rangle/(1 + \sin\vartheta)$, so $\langle u, v\rangle = \langle s, v\rangle - \tau\|v\|^2 = \langle s, v\rangle \sin\vartheta/(1 + \sin\vartheta)$. Next, τ is a root of $\tau^2 - 2\rho\tau + r_2 = 0$. Hence $\|u\|^2 = \|s\|^2 - 2\tau\langle s, v\rangle + \tau^2\|v\|^2 = 2\tau\big(\|s\|^2\|v\|^2 - \langle s, v\rangle^2\big)/\langle s, v\rangle = 2\tau\|s\|^2\|v\|^2\sin^2\vartheta/\langle s, v\rangle$. Substituting gives the cosine.

Item 3. $Pv = \tau v + u = s$. P equals τI on $u^\perp$ and has eigenvalue $\tau + \|u\|^2/\langle u, v\rangle$ on u. By item 2,

$$\frac{\|u\|^2}{\langle u, v\rangle} = \frac{2\tau\|s\|^2\|v\|^2\sin\vartheta(1 + \sin\vartheta)}{\langle s, v\rangle^2} = \frac{2\tau\sin\vartheta(1 + \sin\vartheta)}{\cos^2\vartheta} = 2\rho\sin\vartheta,$$

so the eigenvalue on u is $\rho(1 - \sin\vartheta) + 2\rho\sin\vartheta = \rho(1 + \sin\vartheta)$. □

When $\vartheta = 0$ ($v \parallel s$), $u = 0$ and $P = \rho I$, which is the continuous limit of item 3.

Corollary 3.2 (structure and bounds)**.** *1. **Means.** The two eigenvalues have arithmetic mean $\|s\|^2/\langle s, v\rangle$ (the BB1 step [4]), geometric mean $\|s\|/\|v\|$ and harmonic mean $\langle s, v\rangle/\|v\|^2$ (BB2). ZMSHR1 takes a step at most BB2 on $u^\perp$ and at least BB1 along u.*

*2. **Bounds.** Under the mixing constraints, $\cos^2\vartheta \ge \theta_1/\theta_2$. Hence*

$$\lambda_{\min}(P) \ge \frac{1}{\theta_2\big(1 + \sqrt{1 - \theta_1/\theta_2}\big)} > \frac{1}{2\theta_2}, \qquad \lambda_{\max}(P) \le \frac{1 + \sqrt{1 - \theta_1/\theta_2}}{\theta_1} < \frac{2}{\theta_1},$$
$$\kappa(P) = \frac{1 + \sin\vartheta}{1 - \sin\vartheta} \le \frac{4\theta_2}{\theta_1}.$$

None of these depends on δ_{sg}.

Proof. Item 1 is a direct computation from Lemma 3.1. For item 2, multiply the two mixing constraints: $\cos^2\vartheta = \frac{\langle s,v\rangle}{\|s\|^2}\cdot\frac{\langle s,v\rangle}{\|v\|^2} \geq \theta_1/\theta_2$. Then use $\lambda_{\max} = \rho(1+\sin\vartheta)$ with $\rho \leq 1/\theta_1$, and $\lambda_{\min} = \frac{\langle s,v\rangle}{\|v\|^2(1+\sin\vartheta)}$ with $\frac{\langle s,v\rangle}{\|v\|^2} \geq 1/\theta_2$. □

Corollary 3.3 (the safeguard is inert)**.** *Whenever the mixing step succeeds,* $\cos\angle(u,v) \geq \frac{1}{2}\cos\vartheta \geq \frac{1}{2}\sqrt{\theta_1/\theta_2}$*. Hence the safeguard can only fire if* $\delta_{\mathrm{sg}} \geq \frac{1}{2}\sqrt{\theta_1/\theta_2}$*, or in the degenerate case* $u = 0$*, where it changes nothing. For the published parameters the threshold is* $5\cdot 10^{-5}$*, while* $\delta_{\mathrm{sg}} = 10^{-8}$*.*

The bound $P_k \preceq \frac{\delta_{\mathrm{sg}}+1}{\delta_{\mathrm{sg}}\theta_1} I$ of [20, Lemma 1] is therefore an artefact of a worst-case use of the safeguard. For the published parameters it equals $5.0\cdot 10^{13}$, while Corollary 3.2 gives $\lambda_{\max}(P_k) \leq 1.0\cdot 10^{6}$.

Theorem 3.4 (stability, sharpened)**.** *Assume the hypotheses of [20, Thm. 2], with* $q := (1+\epsilon)\|I - \alpha P_* H\| < 1$ *(cf. Proposition 2.2; q equals the original q if* P_* *is a multiple of* I*). Let*

$$\beta := \sup_k \|P_k - P_*\| \leq \max\left(\tfrac{1+\sqrt{1-\theta_1/\theta_2}}{\theta_1},\, \lambda_*\right).$$

Then

$$\|x_{k+1} - x_*\| \leq q^k\|x_1 - x_*\| + (1+\epsilon)\alpha\sum_{j=1}^{k} q^{k-j}\|(P_j - P_*)\nabla f(x_j)\|, \tag{3}$$

$$\limsup_k \|x_k - x_*\| \leq \frac{(1+\epsilon)\alpha\beta}{1-q}\,\limsup_k \|\nabla f(x_k)\|. \tag{4}$$

Moreover, for $P_* = \tau_* I$*,*

$$\|(P_j - \tau_* I)\nabla f(x_j)\| \leq |\tau_j - \tau_*|\,\|\nabla f(x_j)\| + 2\rho_j \sin\vartheta_j\,\|A\hat{u}_j\|\,\|Ax_j - y\|, \qquad \hat{u}_j = u_j/\|u_j\|.$$

Proof. **Inequality** (3)**.** It is the recursion in the proof of [20, Thm. 2], unrolled, with the norm in place of ρ.

Bound on β**.** By Weyl's inequality, the eigenvalues of $P_j - P_*$ lie in $[\lambda_{\min}(P_j) - \lambda_{\max}(P_*),\ \lambda_{\max}(P_j) - \lambda_{\min}(P_*)]$. Since both matrices are positive definite, $\|P_j - P_*\| < \max(\lambda_{\max}(P_j), \lambda_{\max}(P_*))$. Now apply Corollary 3.2. For the first step, $P_1 = I$ and $\theta_1 < 1$.

Inequality (4)**.** It follows from $e_{k+1} \leq qe_k + c_k \Rightarrow \limsup e_k \leq \limsup c_k/(1-q)$.

Last claim. By Lemma 3.1, $P_j - \tau_* I = (\tau_j - \tau_*)I + 2\rho_j \sin\vartheta_j\,\hat{u}_j\hat{u}_j^*$, and $|\langle \hat{u}_j, A^*(Ax_j - y)\rangle| \leq \|A\hat{u}_j\|\|Ax_j - y\|$. □

Remark 3.5.

- **Size of the improvement.** The constant multiplying $(1+\epsilon)\alpha/(1-q)$ drops from $\frac{\delta_{\mathrm{sg}}+1}{\delta_{\mathrm{sg}}\theta_1} + \lambda_* \approx 5\cdot 10^{13}$ to $\max(10^6, \lambda_*)$. The gradient factor drops from $R \geq \|\nabla f(x_1)\|$ to $\limsup\|\nabla f(x_k)\|$, which decays to nearly zero in the experiments of [20, Fig. 7] [CHECK: figure number].

- **Rank-one refinement.** A large eigenvalue along a near-null direction ($\|A\hat{u}\| \approx 0$) costs nothing. Operator-norm bounds cannot see this.

- **Still conditional.** All of this still needs $q < 1$, which Observation 2.1 rules out under undersampling. Section 4 removes that requirement.

4 Restricted contraction

4.1 Secant tangent structure

Assumption 4.1 (secant tangent structure)**.** *Along the iteration* (1)*, for each k there exist*

- *an orthogonal projector Π_k of rank at most d, onto a subspace T_k;*
- *a linear map S_k with $S_k = \Pi_k S_k \Pi_k$ and $\|S_k\| \le 1$;*
- *a remainder r_k,*

such that

$$D(z_k) - D(x^\natural) = S_k(z_k - x^\natural) + r_k, \qquad \|r_k\| \le \eta \|z_k - x^\natural\|.$$

The map S_k is the part of the denoiser's secant that lives on a d-dimensional subspace: a nonexpansive "tangent" action, which may shrink as well as pass. The remainder r_k is leakage: everything the denoiser lets through outside T_k, together with any gain above 1.

Lemma 4.2 (from Jacobians)**.** *Let D be C^1 on the segment $[x^\natural, z_k]$, and let $\bar{J}_k = \int_0^1 J_D(x^\natural + t(z_k - x^\natural))\,dt$ be the secant Jacobian. If $\|\bar{J}_k - S_k\| \le \eta$ for some S_k as in Assumption 4.1, the assumption holds. In particular it holds if $\|J_D(x) - \Pi_0\| \le \eta_0 + L_J\|x - x^\natural\|$ on a ball containing z_k: take $S_k = \Pi_0$ and $\eta = \eta_0 + \frac{L_J}{2}\|z_k - x^\natural\|$.*

Proof. By the mean value theorem, $D(z_k) - D(x^\natural) = \bar{J}_k(z_k - x^\natural)$. For the second claim, $\|\bar{J}_k - \Pi_0\| \le \int_0^1 (\eta_0 + L_J t\|z_k - x^\natural\|)\,dt$. □

Example 4.3 (soft thresholding: $\eta = 0$ globally)**.** Let W be orthonormal and $D(z) = W^* \mathrm{soft}_\lambda(Wz)$, with some coefficients left unthresholded.

- **Secant slopes.** For scalars $a \ne b$, $\big(\mathrm{soft}_\lambda(a) - \mathrm{soft}_\lambda(b)\big)/(a-b) \in [0,1]$, and the slope vanishes iff $|a|, |b| \le \lambda$.
- **The assumption holds with $\eta = 0$.** Hence $D(z) - D(x^\natural) = S(z - x^\natural)$ with $S = W^* \,\mathrm{diag}(\text{slopes})W$, $\|S\| \le 1$, and $S = \Pi S \Pi$. Here Π projects onto the atoms with $|(Wz)_i| > \lambda$ or $|(Wx^\natural)_i| > \lambda$ (plus the unthresholded ones). So Assumption 4.1 holds with $\eta = 0$ for every z, with d_k the size of that union.
- **Hard thresholding** is discontinuous, so its secant slopes are unbounded near the threshold. Inside the tube where $\big||(Wz_k)_i| - \lambda\big|$ is bounded below and the truth's coefficients are separated from λ, it is exactly the projector onto the truth's support ($\eta = 0$, $L_J = 0$).
- **Learned denoisers.** For trained (bias-free, normalization-equivariant) CNNs the Jacobian is approximately symmetric, with eigenvalues concentrated near 0 and 1, and acts as an adaptive projection whose rank grows as the noise level decreases [32, 24]. Assumption 4.1 quantifies how close this is to a projector; it is a hypothesis to be measured, not a theorem (Section 8).

4.2 The deterministic theorem

Definition 4.4 (restricted constants)**.** With T_k, S_k as in Assumption 4.1, $\varepsilon := D(x^\natural) - x^\natural$ and $\varepsilon_0 := \|\varepsilon\|$, set

$$\delta := \sup_{k\geq 1}\|\Pi_k M_k \Pi_{k-1}\|, \quad m := \sup_k \|M_k\|, \quad W := \sup_k \|\Pi_k w_k\|, \quad W' := \sup_k \|w_k\|, \quad \mu := \sup_k \|\Pi_k M_k \varepsilon\|.$$

- δ is a pairwise restricted isometry constant of the data step between consecutive tangent spaces.
- For $P_k = I$ and $\alpha \leq 1/L$, $m \leq 1$, and $\delta \leq 1 - \alpha \inf \lambda_{\min}(\Pi H \Pi|_T)$ when T_k does not change. For dynamic P_k, M_k is self-adjoint in the P_k^{-1} inner product, so

  $$m \leq \sup_k \sqrt{\kappa(P_k)}\, \max\left(1, \alpha \lambda_{\max}(P_k^{1/2} H P_k^{1/2}) - 1\right),$$

  and Lemma 3.1 gives $\kappa(P_k)$ explicitly.
- $\mu \leq m\varepsilon_0$, but μ is much smaller when the data step carries little of the mismatch into the tangent spaces.

Theorem 4.5 (restricted contraction to the truth)**.** *Suppose Assumption 4.1 holds. Let* $a_k := S_{k-1}(z_{k-1} - x^\natural) = x_k - D(x^\natural) - r_{k-1} \in T_{k-1}$*, and set* $A_k = \|a_k\|$ *and* $E_k = \|x_k - x^\natural\|$*. Then for* $k \geq 1$*:*

1. ***(coupled recursion)***

$$A_{k+1} \leq \delta A_k + \eta m^2 E_{k-1} + W + \mu + \eta m W', \tag{5}$$
$$E_{k+1} \leq A_{k+1} + \eta m E_k + \eta W' + \varepsilon_0; \tag{6}$$

2. ***(scalar recursion)***

$$E_{k+1} \leq (\delta + \eta m) E_k + \eta m(\delta + m) E_{k-1} + W + (1 + \delta + m)(\eta W' + \varepsilon_0). \tag{7}$$

3. ***(convergence)*** *If*

$$\eta m(1 - \delta + m) < 1 - \delta, \tag{8}$$

then the error is dominated by a sequence that converges linearly, with rate

$$r = \tfrac{1}{2}\left(\delta + \eta m + \sqrt{(\delta + \eta m)^2 + 4\eta m(m - \delta)}\right) < 1,$$

and

$$\limsup_{k\to\infty} \|x_k - x^\natural\| \leq \frac{W + \mu + (1 - \delta)\varepsilon_0 + (1 - \delta + m)\,\eta W'}{1 - \delta - \eta m(1 - \delta + m)}. \tag{9}$$

4. ***(no leakage)*** *If* $\eta = 0$*, then* $A_{k+1} \leq \delta A_k + W + \mu$*,* $r = \delta$*, and* $\limsup_k \|x_k - x^\natural\| \leq \varepsilon_0 + \frac{W+\mu}{1-\delta}$.

Proof. By Assumption 4.1 and (2), $e_{k+1} = D(z_k) - D(x^\natural) + \varepsilon = a_{k+1} + r_k + \varepsilon$, with $a_{k+1} = S_k(M_k e_k + w_k)$ and $\|r_k\| \leq \eta(m E_k + W')$. Since $a_k \in T_{k-1}$ and $S_k = S_k \Pi_k$,

$$S_k M_k a_k = S_k\, \Pi_k M_k \Pi_{k-1}\, a_k, \qquad \|S_k M_k a_k\| \leq \delta A_k.$$

Write $e_k = a_k + r_{k-1} + \varepsilon$. Then

$$a_{k+1} = S_k M_k a_k + S_k M_k r_{k-1} + S_k M_k \varepsilon + S_k w_k.$$

- **The four terms.** $\|S_k M_k r_{k-1}\| \le m\eta(mE_{k-1} + W')$, $\|S_k M_k \varepsilon\| \le \|\Pi_k M_k \varepsilon\| \le \mu$, and $\|S_k w_k\| \le \|\Pi_k w_k\| \le W$. Together these give (5).
- **Inequality** (6) is the triangle inequality applied to $e_{k+1} = a_{k+1} + r_k + \varepsilon$.
- **Inequality** (7)**.** Bound instead $A_k \le E_k + \|r_{k-1}\| + \varepsilon_0$ and $\|S_k M_k(r_{k-1}+\varepsilon)\| \le m(\|r_{k-1}\|+\varepsilon_0)$, then insert $\|r_{k-1}\| \le \eta(mE_{k-1} + W')$.
- **Convergence.** The vector $X_k = (A_k, E_k, E_{k-1})$ satisfies $X_{k+1} \le GX_k + c$ componentwise, with

$$G = \begin{pmatrix} \delta & 0 & \eta m^2 \\ \delta & \eta m & \eta m^2 \\ 0 & 1 & 0 \end{pmatrix} \ge 0, \qquad c = \big(c_A,\ c_A + c_E,\ 0\big),$$

  where $c_A = W + \mu + \eta m W'$ and $c_E = \eta W' + \varepsilon_0$. The characteristic polynomial of G is $\lambda\big(\lambda^2 - (\delta + \eta m)\lambda - \eta m(m - \delta)\big)$, and $m \ge \delta$ because $\|\Pi_k M_k \Pi_{k-1}\| \le \|M_k\|$. So $\rho(G) = r$, and $r < 1$ iff $1 - (\delta + \eta m) - \eta m(m - \delta) > 0$, which is (8).
- **The limit.** Since $G \ge 0$ and $\rho(G) < 1$, induction gives $X_k \le G^k X_0' + \sum_{j<k} G^j c$, where X_0' collects the initial values. The right-hand side converges linearly to $(I - G)^{-1}c \ge 0$. Its second component solves $A = \delta A + \eta m^2 E + c_A$ and $E = A + \eta m E + c_E$, i.e. $E = \big(c_A + (1-\delta)c_E\big)/\big(1 - \delta - \eta m(1 - \delta + m)\big)$, which is (9).
- **The case** $\eta = 0$ is immediate.

□

Remark 4.6 (reading the theorem).

1. **What contracts.** Contraction is needed only between consecutive tangent spaces ($\delta < 1$), not on $\mathbb{R}^n$. For PnP-ISTA ($m = 1$) with an exact tangent denoiser ($\eta = 0$), the requirement is $\delta < 1$: a restricted condition on pairs of d-dimensional subspaces, attainable with undersampled Fourier data (Section 5), unlike $\rho(I - \alpha H) < 1$.
2. **Noise.** W is the back-projected noise seen by a d-dimensional subspace. For white noise, $W \approx \alpha\sigma\sqrt{d\,\|\Pi H \Pi\|}$, as in classical compressed sensing. The full noise W' enters only multiplied by the leakage.
3. **Leakage tolerance.** For $m = 1$, (8) reads $\eta < \frac{1-\delta}{2-\delta}$. Without density compensation $1 - \delta \approx 1/\kappa_T$, the restricted condition number of H, so the denoiser's leakage must be below roughly $1/\kappa_T$. A preconditioner raises $1 - \delta$ but may raise m; the trade-off is exactly $\eta m(1 - \delta + m) < 1 - \delta$. Preconditioning thus matters for the *validity* of recovery, not just for speed.
4. **No fixed point needed.** The theorem needs no fixed point, no global Lipschitz constant and no convergence of P_k. It bounds the distance to the truth. It specialises to the local linear convergence of iterative soft thresholding once the support is identified [11, 17, 29], and extends it to non-proximal denoisers, leakage and preconditioning.
5. **Comparison with [30, 23].** Those results require a contraction of $I - \alpha PH$, or a restricted isometry, on the full secant set $\Sigma - \Sigma$ together with a Lipschitz property of D [CHECK: precise statement]. Here the data step is only ever evaluated between tangent spaces of the iterates.

4.3 The exact local rate

Proposition 4.7 (local rate)**.** *Let P be fixed, x_* a fixed point of $T(x) = D(x - \alpha P \nabla f(x))$, and D differentiable at $z_* = x_* - \alpha P \nabla f(x_*)$. Then $T'(x_*) = J_D(z_*)(I - \alpha P H)$. If $\rho(T'(x_*)) < 1$, then x_* is a point of attraction and $\limsup_k \|x_k - x_*\|^{1/k} \le \rho\big(J_D(z_*)(I - \alpha P H)\big)$ [33, 10.1.3–10.1.4]. For soft thresholding at a nondegenerate fixed point, $J_D(z_*) = \Pi_*$ and, for $P = I$ and $\alpha \le 1/L$, $\rho = \|\Pi_*(I - \alpha H)\Pi_*\| = 1 - \alpha\lambda_{\min}(\Pi_* H \Pi_*|_{T_*})$.*

The bound of [20] is an upper bound for this rate, through $\rho(JM) \le \|J\|\|M\|$, and only when $\|M\| = \rho(M)$. In compressed sensing $\rho(M) = 1$, while $\rho(JM) < 1$ is typical. Both quantities are measurable with Jacobian-vector products (power iteration on $w \mapsto J_D(z_*)Mw$).

5 Sampling theorems

Shot model.

- **Shots.** Data come in N_b shots: a readout, a segment or one k-space location, seen by all coils after noise whitening. Shot j has a block operator $B(\omega_j) : \mathbb{R}^n \to \mathbb{R}^b$, with $\omega_1, \dots, \omega_{N_b}$ drawn i.i.d. from a distribution ν. Then $H = \sum_j B(\omega_j)^* B(\omega_j)$, and $\bar{H} := \mathbb{E}_\omega B(\omega)^* B(\omega)$ satisfies $\mathbb{E}H = N_b \bar{H}$. Deterministic shots, such as a fully sampled centre, can be added at no cost.
- **Tangent at the truth.** T_0 is a d-dimensional subspace with projector Π_0. It is a property of the denoiser and $x^\natural$, not of the sampling.
- **Restricted expected spectrum:** $\lambda_- = \lambda_{\min}(\Pi_0 \bar{H} \Pi_0|_{T_0})$ and $\lambda_+ = \lambda_{\max}(\Pi_0 \bar{H} \Pi_0|_{T_0})$.
- **Block coherence.** $\Phi_0 = \operatorname{ess\,sup}_\omega \|B(\omega)\Pi_0\|^2$. Since $\mathbb{E}\|B\Pi_0\|_F^2 = \operatorname{tr} \Pi_0 \bar{H} \Pi_0 \ge d\lambda_-$ and $\operatorname{rank} B \le b$, we have $\Phi_0 \ge d\lambda_- / \min(d, b)$. Write $\Phi_0 = \kappa_{\mathrm{coh}}\, d\lambda_-/b$ when $d > b$, where $\kappa_{\mathrm{coh}} \ge 1$ measures how unevenly shots see T_0.

Assumption 5.1 (tangent structure at the truth)**.** *D is C^1 on $\bar{B}(x^\natural, R)$, and $\|J_D(x) - \Pi_0\| \le \eta_0 + L_J\|x - x^\natural\|$ there.*

Theorem 5.2 (local, non-uniform recovery)**.** *Let Assumption 5.1 hold, $P = I$ and $\alpha = 1/\|H\|$. Let $t, p \in (0, 1)$ and*

$$N_b \ge \frac{3\,\Phi_0}{t^2 \lambda_-} \log \frac{2d}{p}. \tag{10}$$

Then with probability at least $1 - p$ over the shots:

(a) $(1 - t)N_b\lambda_- \le \lambda_{\min}(\Pi_0 H \Pi_0|_{T_0}) \le \lambda_{\max}(\Pi_0 H \Pi_0|_{T_0}) \le (1 + t)N_b\lambda_+$, *hence*

$$\delta_0 := \|\Pi_0(I - \alpha H)\Pi_0\| \le 1 - \frac{(1 - t)N_b\lambda_-}{\|H\|}.$$

(b) Let $w = \alpha A^* n$, $W = \|\Pi_0 w\|$, $W' = \|w\|$, $\mu_0 = \|\Pi_0(I - \alpha H)\varepsilon\|$ and $\eta(r) = \eta_0 + \frac{L_J}{2}(r + W')$. Suppose $r > 0$ satisfies $r + W' \le R$, $c(r) := \delta_0 + \eta(r)(2 + \delta_0) < 1$ and $W + (2 + \delta_0)\big(\eta(r)W' + \varepsilon_0\big) \le (1 - c(r))\, r$. Then for every initialisation with $\|x_0 - x^\natural\| \le r$ and $\|x_1 - x^\natural\| \le r$, all iterates stay in $\bar{B}(x^\natural, r)$, and

$$\limsup_k \|x_k - x^\natural\| \le \frac{W + \mu_0 + (1 - \delta_0)\varepsilon_0 + (2 - \delta_0)\eta(r)W'}{1 - \delta_0 - (2 - \delta_0)\eta(r)},$$

approached at a linear rate.

(c) On the event of (a), with probability at least $1 - e^{-u}$ over the noise,

$$W \le \alpha\sigma\sqrt{(1+t)N_b\lambda_+}\,\big(\sqrt{d} + \sqrt{2u}\big), \qquad W' \le \alpha\sigma\big(\sqrt{\operatorname{tr} H} + \sqrt{2u\|H\|}\big).$$

Proof. **(a) Matrix Chernoff.** The matrices $X_j = \Pi_0 B(\omega_j)^* B(\omega_j)\Pi_0|_{T_0}$ are independent, positive semidefinite and $d \times d$, with $\lambda_{\max}(X_j) \le \Phi_0$ and $\sum_j \mathbb{E}X_j = N_b\Pi_0\bar{H}\Pi_0|_{T_0}$. The matrix Chernoff inequality [42, Thm. 1.1] gives

$$\Pr\big[\lambda_{\min}(\textstyle\sum X_j) \le (1-t)N_b\lambda_-\big] \le d\, e^{-t^2 N_b\lambda_-/(2\Phi_0)},$$
$$\Pr\big[\lambda_{\max}(\textstyle\sum X_j) \ge (1+t)N_b\lambda_+\big] \le d\, e^{-t^2 N_b\lambda_+/(3\Phi_0)}.$$

By (10) and $\lambda_+ \ge \lambda_-$, each probability is at most $p/2$. Since $0 \preceq \alpha H \preceq I$, the compression $\Pi_0(I - \alpha H)\Pi_0|_{T_0}$ has its eigenvalues in $[0,\, 1 - \alpha\lambda_{\min}(\Pi_0 H \Pi_0|_{T_0})]$.

(b) Apply Theorem 4.5 with $\Pi_k = S_k = \Pi_0$ and $m = 1$; the pairwise constant is then δ_0.

- **Leakage inside the tube.** By Lemma 4.2, $\|r_k\| \le \eta_k\|z_k - x^\natural\|$ with $\eta_k = \eta_0 + \frac{L_J}{2}\|z_k - x^\natural\|$, provided $z_k \in \bar{B}(x^\natural, R)$.
- **Invariance.** If $E_j \le r$ for $j \le k$, then $\|z_j - x^\natural\| \le E_j + W' \le R$, so $\eta_j \le \eta(r)$. The scalar recursion (7) then gives $E_{k+1} \le c(r)r + W + (2+\delta_0)(\eta(r)W' + \varepsilon_0) \le r$.
- **Limit.** Within the tube the coupled recursion applies with $\eta = \eta(r)$. Its condition $(2-\delta_0)\eta < 1 - \delta_0$ is implied by $c(r) < 1$, and (9) gives the bound.

(c) Gaussian concentration. $\Pi_0 A^* n \sim \mathcal{N}(0, \sigma^2\Pi_0 H \Pi_0)$. Its norm is a $\sigma\|\Pi_0 H\Pi_0\|^{1/2}$-Lipschitz function of a standard Gaussian vector, with mean at most $\sigma(\operatorname{tr}\Pi_0 H\Pi_0)^{1/2}$ [10]. Use $\operatorname{tr}\Pi_0 H\Pi_0 \le d\,\lambda_{\max}$ and the upper bound in (a). The same argument applies to A^*n. □

Remark 5.3 (reading Theorem 5.2). 1. **Leading behaviour.** With $\eta \to 0$ and $\alpha = 1/\|H\|$,

$$\limsup_k \|x_k - x^\natural\| \lesssim \varepsilon_0 + \frac{\sigma}{\sqrt{N_b}} \cdot \frac{\sqrt{(1+t)\,d\,\lambda_+}}{(1-t)\lambda_-} + \frac{\|H\|}{(1-t)N_b\lambda_-}\,\mu_0.$$

The noise term is the classical compressed-sensing term: $\sqrt{d}$, decaying as $N_b^{-1/2}$, times the restricted conditioning.

2. **The mismatch term.** The mismatch enters through $\mu_0 = \|\Pi_0(I - \alpha H)\varepsilon\|$. For a mismatch orthogonal to T_0, $\mu_0 = \alpha\|\Pi_0 H\varepsilon\|$. Its mean $\alpha N_b\Pi_0\bar{H}\varepsilon$ vanishes when $\bar{H}$ is isotropic. A peaked sampling density (no density compensation) couples $T_0^\perp$ into T_0 and amplifies the mismatch by the restricted condition number.

3. **Sample complexity.** If $d > b$, (10) reads: total samples $N_b b \gtrsim \kappa_{\text{coh}}\, d \log(2d/p)/t^2$. It is proportional to the local Jacobian rank, as in the Gaussian-mixture phase transition [38], and not to the size of a model union.

4. **The moving basis does not matter locally.** Inside the tube every secant Jacobian is within $\eta(r)$ of the fixed projector Π_0. A restricted isometry is needed on a single d-dimensional subspace, and the guarantee is non-uniform (for this $x^\natural$, with high probability over the shots). The price is the tube condition, roughly $r \lesssim (1-\delta_0)/(L_J(2+\delta_0))$: the warm start must already be within the curvature radius of the denoiser. This is the theoretical role of initialisation.

Remark 5.4 (deterministic shots and the coherence barrier). Φ_0 is a worst case over shots. For wavelet-type tangent spaces and variable-density sampling, it is dominated by the centre of k-space, where coarse-scale atoms concentrate their energy. This is the coherence barrier of [2], and with it (10) becomes very conservative (Section 7). Two remedies:

1. **Deterministic shots.** Shots that are sampled deterministically, such as a fully sampled centre, contribute a fixed positive semidefinite $C_0 = \Pi_0 H_{\mathrm{det}} \Pi_0$. The summands $X_j + C_0/N_b$ are again independent and positive semidefinite, with sum $\Pi_0 H \Pi_0$. The Chernoff step then holds with Φ_0 replaced by $\Phi_{\mathrm{rand}} + \|C_0\|/N_b$ (coherence over the random shots only), and with $N_b\lambda_-$ replaced by $\lambda_{\min}(C_0 + N_b \Pi_0 \bar{H}_{\mathrm{rand}} \Pi_0|_{T_0})$.
2. **Multilevel analysis.** Split T_0 by scale, as in [2].

[TODO: remedy 1 in the toy.]

Theorem 5.5 (uniform over a tangent family)**.** ***Setting.***

- *$\mathcal{T}$ is a family of subspaces of dimension d, and $\mathcal{N}(\mathcal{T}, \epsilon)$ is its covering number in projector distance, with centres in $\mathcal{T}$.*
- *$\Phi_{\mathcal{T}} = \operatorname{ess\,sup}_\omega \sup_{T \in \mathcal{T}} \|B(\omega)\Pi_T\|^2$ and $\lambda_+(\mathcal{T}) = \sup_T \lambda_{\max}(\Pi_T \bar{H} \Pi_T)$.*
- *$\alpha > 0$ is deterministic, $a := \alpha N_b$ satisfies $a\lambda_{\max}(\bar{H}) \le 1$, and the event $\{\|\alpha H\| \le 2\}$ holds.*

Statement. *For $t \in (0, 1]$, if*

$$N_b \ge \frac{8a\Phi_{\mathcal{T}}\,(a\lambda_+(\mathcal{T}) + t/3)}{t^2} \log \frac{2d\,\mathcal{N}(\mathcal{T}, t/12)^2}{p},$$

then with probability at least $1 - p$, simultaneously for all $T, T' \in \mathcal{T}$,

$$\|\Pi_T (I - \alpha H)\Pi_{T'}\| \le \|\Pi_T (I - a\bar{H})\Pi_{T'}\| + t.$$

Consequently Theorem 4.5 applies with $\delta \le \bar{\delta}_{\mathcal{T}} + t$, where $\bar{\delta}_{\mathcal{T}} := \sup_{T,T'} \|\Pi_T (I - a\bar{H})\Pi_{T'}\|$, to every *iteration whose tangent spaces T_k stay in $\mathcal{T}$.*

Proof. See Appendix A. □

Remark 5.6 (what remains open).

- **Two halves.** Theorem 5.5 is the probabilistic half of the moving-basis problem. The deterministic half is that a trained denoiser has secant tangent structure with small leakage *far* from the truth.
- **Long segments.** Along a long segment the secant averages the projectors of all the tangent spaces it crosses. For a curved model set this gives $\eta \sim \|z - x^\natural\|/\mathrm{reach}$, which brings back the tube of Theorem 5.2.
- **Exceptions.** Piecewise-linear denoisers whose pieces are projectors (soft thresholding is one, Example 4.3), and curvature-corrected arguments as in [44].
- **Price of uniformity.** Relative to (10), Theorem 5.5 pays an extra factor of order $\lambda_{\max}(\bar{H})/\lambda_-$, because cross terms $T \ne T'$ need an additive rather than a relative deviation bound.

6 Preconditioners: the estimator and the design target

6.1 The PnP limit depends on P

Consider the local affine model $D(z) = \Pi_0 z + c$, with $c \in \mathbb{R}^n$ arbitrary, and a fixed $P \succ 0$. Soft thresholding has this form near any z whose coefficients stay away from $\pm\lambda$; there $\Pi_0 c$ is the shrinkage. The mismatch is $\varepsilon = D(x^\natural) - x^\natural = c - \Pi_0^\perp x^\natural$, with $\Pi_0 \varepsilon = \Pi_0 c$. Let $G_P := \Pi_0 P H \Pi_0|_{T_0} : T_0 \to T_0$, and let Q_0 be the orthogonal projector onto $\operatorname{range}(A\Pi_0)$ in data space.

Proposition 6.1 (preconditioning changes the estimator)**.** *Assume $\Pi_0 H \Pi_0|_{T_0}$ and G_P are invertible.*

1. ***Fixed point.*** *PnP has the unique fixed point $\hat{x}_P = x^\natural + b_P + K_P n$, with $K_P = G_P^{-1}\Pi_0 P A^*$ and bias*
$$b_P = \Pi_0^\perp \varepsilon + (\alpha G_P)^{-1}\Pi_0 \varepsilon - G_P^{-1}\Pi_0 P H \Pi_0^\perp \varepsilon. \tag{11}$$
The iteration converges to it iff $\rho(I - \alpha G_P) < 1$.

2. ***Gauss–Markov.*** *$K_P A \Pi_0 = \Pi_0$, so the noise part is unbiased on T_0, and*
$$\operatorname{Cov}(\hat{x}_P) = \sigma^2 K_P K_P^* = \sigma^2 (\Pi_0 H \Pi_0)^{-1} + \sigma^2\, G_P^{-1}\Pi_0 P A^* (I - Q_0) A P \Pi_0 G_P^{-*},$$
where the first term is the covariance of the best linear unbiased estimator (BLUE) $K_* = (\Pi_0 H \Pi_0)^{-1}\Pi_0 A^*$.

3. ***Criterion.*** *$K_P = K_*$ iff $\Pi_0 P A^*(I - Q_0) = 0$, i.e. iff P maps $V_0 := T_0^\perp \cap (\operatorname{null} A)^\perp$ into $T_0^\perp$. A sufficient condition is $P T_0 \subseteq T_0$. If A is injective, $V_0 = T_0^\perp$ and invariance $P T_0 \subseteq T_0$ is also necessary.*

4. ***Bias.*** *The in-tangent mismatch $\Pi_0 \varepsilon$ is amplified by $(\alpha G_P)^{-1}$. For $P = I$ and $\alpha = 1/L$ this factor has norm $L/\lambda_{\min}(\Pi_0 H \Pi_0|_{T_0}) = \kappa_{T_0}$. It equals the identity iff $\alpha G_P = I_{T_0}$. The off-tangent mismatch leaks into T_0 through $G_P^{-1}\Pi_0 P H \Pi_0^\perp$.*

Proof. **Item 1.** A fixed point satisfies $\hat{x} = \Pi_0\big(\hat{x} - \alpha P A^*(A\hat{x} - y)\big) + c$. Hence $\Pi_0^\perp \hat{x} = \Pi_0^\perp c$, so $\Pi_0^\perp(\hat{x} - x^\natural) = \Pi_0^\perp \varepsilon$. Also $\alpha \Pi_0 P A^*(A\hat{x} - y) = \Pi_0 c = \Pi_0 \varepsilon$. With $e = \hat{x} - x^\natural = e_T + \Pi_0^\perp \varepsilon$ and $A\hat{x} - y = Ae - n$, this reads $G_P e_T = \Pi_0 P A^* n + \alpha^{-1}\Pi_0 \varepsilon - \Pi_0 P H \Pi_0^\perp \varepsilon$, which gives (11). After one step the iterates satisfy $\Pi_0^\perp x_k = \Pi_0^\perp c$, and the tangent component follows the affine recursion $e_{T,k+1} = (I - \alpha G_P) e_{T,k} + \text{const}$.

Item 2. $K_P A \Pi_0 = G_P^{-1}\Pi_0 P H \Pi_0 = I_{T_0}$. Next,
$$K_P - K_* = G_P^{-1}\big(\Pi_0 P A^* - \Pi_0 P H \Pi_0 (\Pi_0 H \Pi_0)^{-1}\Pi_0 A^*\big) = G_P^{-1}\Pi_0 P A^*(I - Q_0),$$
using $H = A^*A$ and $Q_0 = A\Pi_0(\Pi_0 H \Pi_0)^{-1}\Pi_0 A^*$. The cross term vanishes, $K_*(K_P - K_*)^* = 0$, because $\Pi_0 A^*(I - Q_0) = 0$. Hence $K_P K_P^* = K_* K_*^* + (K_P - K_*)(K_P - K_*)^*$, and $K_* K_*^* = (\Pi_0 H \Pi_0)^{-1}$.

Item 3. By item 2, $K_P = K_*$ iff $\Pi_0 P A^*(I - Q_0) = 0$. The range of $A^*(I - Q_0)$ is $A^*\big(\operatorname{range}(AT_0)^\perp\big)$, which equals $T_0^\perp \cap \operatorname{range} A^* = V_0$: if $v = A^* q$ with $q \perp AT_0$, then $\langle t, v\rangle = \langle At, q\rangle = 0$ for $t \in T_0$; conversely, $A^* q \perp T_0$ forces $q \perp AT_0$. If $P T_0 \subseteq T_0$ then $P T_0^\perp \subseteq T_0^\perp$ by symmetry. If A is injective, $V_0 = T_0^\perp$.

Item 4. For $P = I$ and $\alpha = 1/L$, $(\alpha G_I)^{-1} = L(\Pi_0 H \Pi_0|_{T_0})^{-1}$. □

Corollary 6.2 (consequences)**.** *1.* ***Scalar.*** *$P = I$, and any scalar P, give the BLUE covariance on the tangent space. Without density compensation, however, they amplify the in-tangent mismatch (for soft thresholding: the shrinkage) by up to κ_{T_0}. This is the familiar bias of the Lasso, $\lambda(A_T^* A_T)^{-1}\mathrm{sign}$.*

2. ***Density compensation.*** *A circulant (density-compensation-type) preconditioner does not respect a signal-adaptive T_0. Its PnP limit has the excess covariance of Proposition 6.1, and it does not equalise αG_P on T_0 either. Unlike in convex optimisation, the preconditioner changes the answer and not only the path. Density compensation inside the data fit is the same phenomenon: a weighted least-squares fit with weights different from the noise precision [36, 37].*

3. ***Frozen ZMSHR1.*** *For $P = \tau I + \lambda_u \hat{u}\hat{u}^*$ (as in Lemma 3.1, with $\lambda_u = 2\rho \sin\vartheta$),*

$$\Pi_0 P A^* (I - Q_0) = \lambda_u \, (\Pi_0 \hat{u}) \big((I - Q_0) A \hat{u} \big)^*.$$

So $K_P = K_*$ iff $\hat{u} \perp T_0$ or $A\hat{u} \in AT_0$. Since $\hat{u} \propto s - \tau H s$, a secant $s \in T_0$ does not *give $\hat{u} \in T_0$: Hs leaves the tangent space. The excess is rank one, of size $\sigma^2 \lambda_u^2 \|(I - Q_0)A\hat{u}\|^2 \|G_P^{-1} \Pi_0 \hat{u}\|^2$.*

4. ***Tangent-restricted.*** *$P_T = \Pi_0 (\Pi_0 H \Pi_0)^{-1} \Pi_0 + \tau(I - \Pi_0)$ leaves T_0 invariant, so its noise covariance is the BLUE's. At $\alpha = 1$, $\alpha G_{P_T} = I_{T_0}$, so the in-tangent mismatch is not amplified, and the off-tangent leak $G_P^{-1} \Pi_0 P H \Pi_0^\perp \varepsilon = (\Pi_0 H \Pi_0)^{-1} \Pi_0 H \Pi_0^\perp \varepsilon$ is the same as for $P = I$. Moreover $\delta(P_T) = \|\Pi_0 (I - P_T H) \Pi_0\| = 0$: the tangent component converges in one step. In the local model, P_T thus combines the variance of $P = I$ with the bias of an ideal preconditioner. Its full-space norm $m = \|I - P_T H\|$ can be large, which by (8) lowers the tolerated leakage.*

5. ***Non-convergence.*** *Even with an exact projector denoiser, PnP with $P \succ 0$ fails to converge if G_P has an eigenvalue with nonpositive real part, which can happen for $n \geq 2$ (Proposition 2.2).*

Remark 6.3 (dynamic preconditioners). For $P_k \to P_\infty$, a limit of the iterates is a fixed point for P_∞, and Proposition 6.1 applies to P_∞. ZMSHR1 depends on the secant only through its direction. Whether P_k converges therefore depends on whether $s_k/\|s_k\|$ converges, typically to the dominant eigenvector of the linearised iteration. A warm start with a different preconditioner does not affect the limit, which is determined by the final P.

6.2 Restricted secants: why tight coefficient bounds slow ZMSHR1

Proposition 6.4 (restricted spectral bound for ZMSHR1)**.** *Let U be a subspace with $\langle w, Hw \rangle \geq (1 - \delta_U) L \|w\|^2$ for $w \in U$, where $L = \|H\|$.*

1. *If $s_k \in U$, $\theta_1 \leq (1 - \delta_U)L$ and $\theta_2 \geq L$, the mixing step accepts $a = 0$ (the pure secant $v = Hs$). Then*

$$\lambda_{\max}(P_k) = \rho_k (1 + \sin\vartheta_k) \leq \frac{1 + \sin\vartheta_k}{(1 - \delta_U)L} < \frac{2}{(1 - \delta_U)L}, \qquad \lambda_{\min}(P_k) \geq \frac{1}{2L},$$

independently of θ_1 and δ_{sg}.

2. *If $s_k = s_U + s_\perp$ with $s_U \in U$, $s_\perp \perp U$, $\|s_\perp\| \leq \xi \|s_k\|$, and $\gamma := \sqrt{(1 - \delta_U)(1 - \xi^2)} - \xi > 0$, the same holds with $(1 - \delta_U)$ replaced by γ^2.*

Proof. **Mixing accepts $a = 0$.** For $a = 0$: $\langle s, Hs\rangle/\|s\|^2 \geq (1-\delta_U)L \geq \theta_1$, and $\|Hs\|^2 \leq L\langle s, Hs\rangle$, so $\|v\|^2/\langle s, v\rangle \leq L \leq \theta_2$.

Item 1. Then $\rho_k = \|s\|^2/\langle s, Hs\rangle \leq 1/((1-\delta_U)L)$, and Lemma 3.1 gives $\lambda_{\max}$. Also $\lambda_{\min} = \frac{\langle s, Hs\rangle}{\|Hs\|^2(1+\sin\vartheta)} \geq \frac{1}{2L}$.

Item 2. $\langle s, Hs\rangle^{1/2} = \|H^{1/2}s\| \geq \|H^{1/2}s_U\| - \|H^{1/2}s_\perp\| \geq \sqrt{(1-\delta_U)L}\,\|s_U\| - \sqrt{L}\,\|s_\perp\|$, with $\|s_U\| = \sqrt{1 - \|s_\perp\|^2/\|s\|^2}\,\|s\|$. □

Remark 6.5 (consequences for parameter choice).

- **Where the secants live.** In the regime of Theorem 4.5, $s_k = a_k - a_{k-1} + (r_{k-1} - r_{k-2})$ lies in $T_{k-1} + T_{k-2}$ up to leakage. For soft thresholding, s_k lies *exactly* in the span of the supports of x_k and x_{k-1} ($\xi = 0$). On such secants ZMSHR1 is automatically bounded by $2/((1-\delta_U)L)$.

- **Where the large eigenvalues come from.** The large eigenvalues that a tiny θ_1 permits arise only for secants with large null-space content. Those are the directions the denoiser removes.

- **Cost of a small worst-case bound.** A small worst-case bound requires raising θ_1 (or δ_{sg} above $\frac{1}{2}\sqrt{\theta_1/\theta_2}$). The mixing step then chooses $a > 0$ whenever the restricted curvature $\langle s, Hs\rangle/\|s\|^2$ is below θ_1, which is exactly when curvature information is most valuable. P_k then tends to ρI with $\rho \leq 1/\theta_1$, and the method loses its curvature adaptation. This is the slowdown reported in [20].

- **The design target** for a preconditioner in PnP is the restricted constant $\|\Pi_T(I - \alpha PH)\Pi_{T'}\|$ on tangent pairs, subject to keeping $m = \|I - \alpha PH\|$ moderate and, for the limit, to the invariance condition of Proposition 6.1. It is not $\rho(I - \alpha PH)$, which is at least 1 in compressed sensing.

7 Numerical illustration

The experiments use a toy problem in which every quantity in the theorems can be computed exactly. They are illustrations, not a benchmark. Code: `experiments.py` (CPU, about half an hour). The numbers in this section are macros generated by `make_numbers.py` from `results/`.

Setup.

- **Phantom and coils.** Real 64×64 modified Shepp–Logan phantom, and 4 smooth complex receive coils.

- **Shots.** A shot is one k-space location seen by all coils ($b = 8$ real values). $N_b = 520$ shots are drawn i.i.d. from $\nu(k) \propto (1 + |k|^2/16^2)^{-1}$, giving 489 distinct locations. There is no density compensation.

- **Null space.** The real data dimension is 3912, and $\operatorname{rank} H = 3500 < n = 4096$, so A has a nontrivial null space.

- **Noise and denoiser.** Gaussian noise at 1% of the data rms. The denoiser is orthonormal Haar soft thresholding (4 levels, coarse band kept, $\lambda = 0.005$), so $\eta = 0$ globally (Example 4.3). The truth's tangent dimension is $d_0 = 737$.

- **Normalisation.** $\|H\| = 1$ and $\alpha = 1$. Iterations start from $D(A^*y)$.

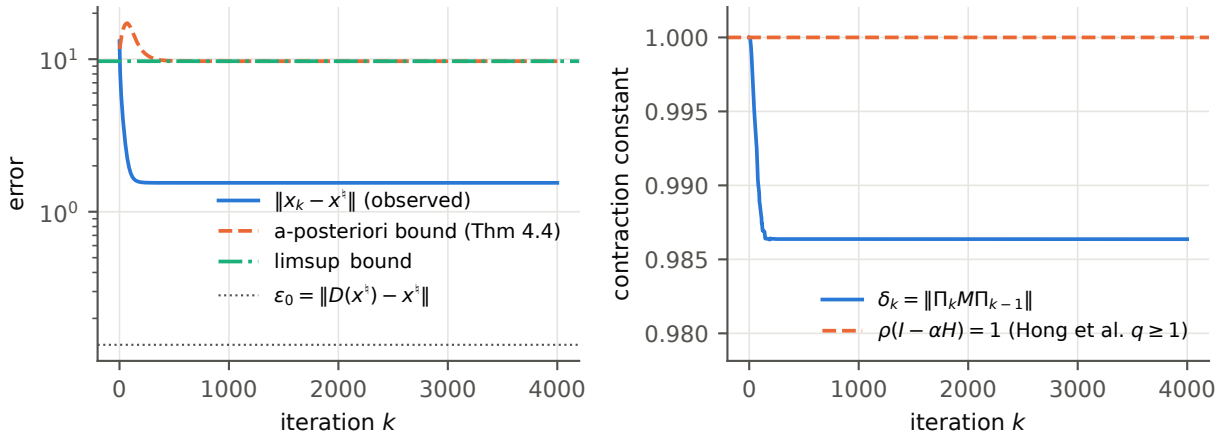


Figure 1: PnP-ISTA with Haar soft thresholding on undersampled multi-coil data. *Left:* error to the truth, the a-posteriori bound of Theorem 4.5(iv), the limsup bound (9) and the mismatch floor ε_0. *Right:* the pairwise restricted constant δ_k against the global quantity $\rho(I - \alpha H) = 1$ that governs [20].

Restricted contraction (Figure 1). PnP-ISTA, 4000 iterations.

- **Hong et al.'s condition fails.** $\lambda_{\min}(H) = 0$, so $q \geq 1$ (Observation 2.1).
- **Restricted constants.** The pairwise constants of Definition 4.4 satisfy $\delta_k < 1$ throughout, with $\sup \delta_k = 0.9864$ over the second half of the run. The secant tangent dimensions range over $d_k \in [899, 3548]$: large in the first iterations, when z_k is dominated by aliasing.
- **The bound holds.** The a-posteriori bound of Theorem 4.5(iv), driven by the measured δ_k, W_k and μ_k, dominates the observed error at every iteration (0 violations).
- **Tightness.** The limsup bound (9) is 9.70 against an observed final error of 1.55, a factor 6.3. The bound uses the worst direction of each $\Pi_k M \Pi_{k-1}$, while the error is spread over many directions.
- **Where the error comes from.** The final error is much larger than the mismatch floor $\varepsilon_0 = 0.134$: the shrinkage bias is amplified by the restricted conditioning (Proposition 6.1(4)).
- **The local rate.** The local rate of Proposition 4.7 at the limit ($d_* = 780$) is 0.9811, against an observed asymptotic rate of 0.9806.

The ZMSHR1 spectrum (Figure 2). P^{2}nP-D with the published parameters:

- **Safeguard.** It fired 0 times. The smallest $\cos \angle(u_k, v_k)$ was 0.140, against the guaranteed $5 \cdot 10^{-5}$ (Corollary 3.3) and $\delta_{\text{sg}} = 10^{-8}$.
- **Mixing.** It chose $a > 0$ in 0 iterations.
- **Largest eigenvalue.** $\max_k \lambda_{\max}(P_k) = 117$, against the δ_{sg}-free bound 10^6 and the published bound $5 \cdot 10^{13}$. It stayed below the restricted bound of Proposition 6.4, with U_k = span of the supports of x_k and x_{k-1}, at every iteration (largest ratio 0.53). That bound is vacuous only in the first iterations, when U_k exceeds the rank of H.

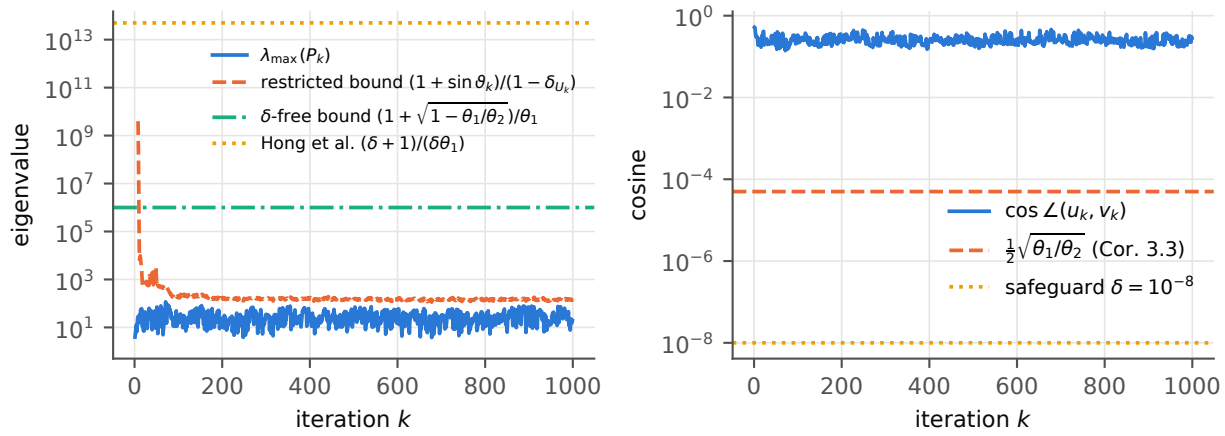


Figure 2: ZMSHR1 along a P^{2}nP-D run. *Left:* $\lambda_{\max}(P_k)$ against the restricted bound of Proposition 6.4, the δ_{sg}-free bound of Corollary 3.2 and the bound of [20, Lemma 1]. *Right:* $\cos\angle(u_k, v_k)$ against the lower bound of Corollary 3.3 and the safeguard threshold.

Table 1: ZMSHR1 with different θ_1 ($\theta_2 = 200$, $\delta_{\mathrm{sg}} = 10^{-8}$). Median error to the truth over the last 20 % of the iterations, fraction of iterations with $a > 0$, and median $\lambda_{\max}(P_k)$.

θ_1	median late error	$a > 0$	median $\lambda_{\max}(P_k)$
$2 \cdot 10^{-6}$	0.48	0 %	25.3
0.01	0.48	0 %	25.3
0.1	0.38	35 %	14.5
0.3	0.81	77 %	5.0
0.6	1.19	99 %	2.0

- **The iteration does not settle.** The error of P^{2}nP-D drops below the PnP-ISTA limit within about 100 iterations, but keeps fluctuating (0.54 at the last iteration) instead of converging to a fixed point. P_k keeps changing with the secant direction.

θ_1 (Figure 3, Table 1).

- **Small θ_1 never binds.** For $\theta_1 \le 10^{-2}$ the mixing step never fires: the restricted curvature $\langle s, Hs\rangle/\|s\|^2$ of the secants stays above θ_1, as Proposition 6.4 predicts.
- **Moderate θ_1 is harmless, large θ_1 is not.** At $\theta_1 = 0.1$ the mixing step fires in about a third of the iterations. It damps the fluctuations of the dynamic iteration, and the late error was the lowest of the sweep. Once θ_1 exceeds the typical restricted curvature (0.3, 0.6), $a > 0$ in most iterations, P_k shrinks towards a scaled identity, and the late error grows towards that of PnP-ISTA (1.55). A small *worst-case* bound on P_k, which needs a large θ_1, thus costs accuracy at a fixed iteration count. Moderate values cost nothing, in line with Proposition 6.4.

The estimator depends on P (Table 2).

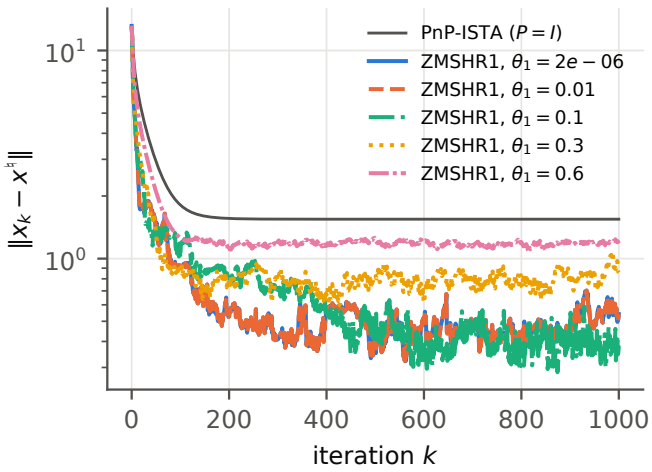


Figure 3: Error to the truth for ZMSHR1 with different θ_1 ($\theta_2 = 200$, $\delta_{\text{sg}} = 10^{-8}$) and for PnP-ISTA.

Table 2: Preconditioning changes the estimator. *Local affine model* (Proposition 6.1): trace of the noise covariance relative to the BLUE, spectral radius of $I - G_P$ on T_0 (the PnP iteration converges iff it is < 1), and squared bias. *Soft-threshold PnP:* Monte Carlo over noise. The ZMSHR1 row uses the final P_k of a P^{2}nP-D run, frozen, in the local model, and the dynamic iteration in the Monte Carlo.

	local affine model			soft-threshold PnP (Monte Carlo)		
P	$\operatorname{tr}\operatorname{Cov}/\operatorname{tr}\operatorname{Cov}_{\text{BLUE}}$	$\rho(I - G_P)$	bias^2	MSE	bias^2	variance
I	1.0000	0.9841	3.628	2.381	2.361	0.0207
circulant (density)	1.1026	0.9849	5.748	3.664	3.642	0.0233
ZMSHR1 (frozen; MC dynamic)	1.0014	0.9848	0.569	0.223	0.095	0.1374
tangent-restricted P_T	1.0000	0.0000	0.018	–	–	–

- **Exact local model.** We take T_0 to be the truth's tangent space ($d_0 = 737$) and the local affine model of soft thresholding at $x^\natural$. The covariances and biases of Proposition 6.1 are then computed exactly.
- **Variance.** The tangent-restricted preconditioner reproduces the BLUE to rounding. The circulant and frozen-ZMSHR1 preconditioners have excess variance.
- **Bias.** The in-tangent shrinkage bias dominates the error. $P = I$ amplifies it by up to κ_{T_0}. P_T does not, and has both the smallest bias and the BLUE variance.
- **Nonlinear Monte Carlo.** Over 16 noise draws we run the actual soft-threshold PnP: with $P = I$ and with the circulant preconditioner until the relative step is below 10^{-9}, and dynamic ZMSHR1 for a fixed number of iterations (it does not converge; see above).

Sample complexity (Figure 4).

- **Setup.** We fix a tangent space, draw N_b shots repeatedly from ν, and record how often the lower Chernoff event of Theorem 5.2(a) holds ($t = 1/2$). Two kinds of tangent space:

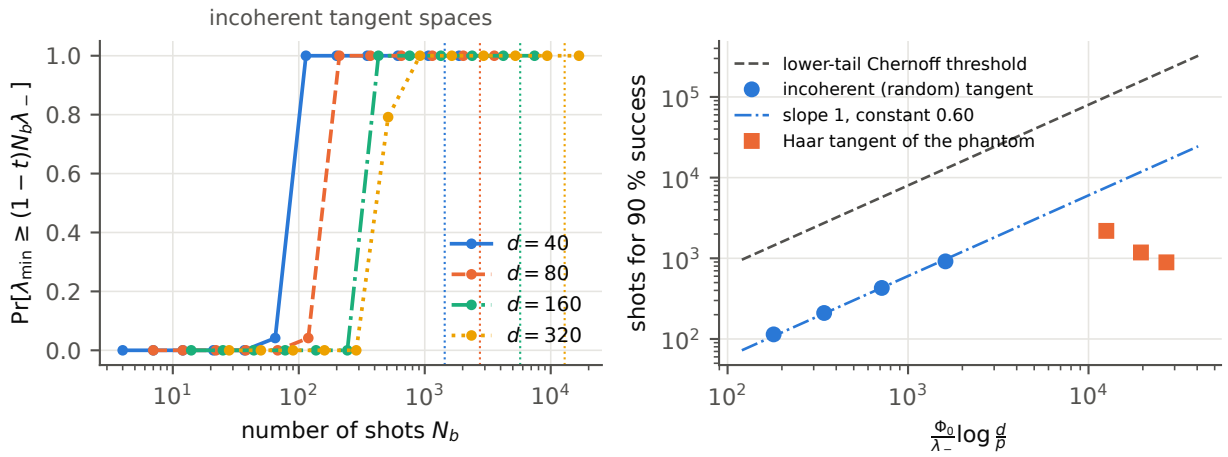


Figure 4: Restricted Chernoff event of Theorem 5.2(a). *Left:* probability versus the number of shots for incoherent tangent spaces; dotted lines mark the lower-tail threshold $\frac{2\Phi_0}{t^2\lambda_-}\log\frac{d}{p}$. *Right:* shots needed for 90 % success against $(\Phi_0/\lambda_-)\log(d/p)$, for incoherent (circles) and Haar (squares) tangent spaces.

  - spans of d random images ($d \in \{40, 80, 160, 320\}$), which are incoherent: $\kappa_{\text{coh}} = 4.8 - -6.0$;
  - Haar tangent spaces of the phantom, i.e. its largest coefficients plus the coarse band ($d \in \{76, 121, 305\}$), with $\kappa_{\text{coh}} = 88 - -199$.

- **Incoherent case.** The empirical 90 % point scales linearly in $(\Phi_0/\lambda_-)\log(d/p)$, as (10) predicts. The theorem's constant is conservative by a factor of 13–14.
- **Haar case.** The points lie far below the prediction (a factor 46–243): the coherence barrier of Remark 5.4. Φ_0 is attained at the centre of k-space, which variable-density sampling covers almost surely anyway.

MRI-scale measurements. [TODO: 4D flow pipeline, phantom ph0001, one frame, trained denoiser:

- $d = \operatorname{tr} J_D$ by Hutchinson / SURE divergence [22, 40];
- Π_0 and η_0 by block Lanczos on $\frac{1}{2}(J + J^*)$;
- κ_{T_0} and Φ_0 per interleaf;
- L_J by finite differences;
- ε_0, and the local rate against the observed rate;
- the excess variance of the circulant preconditioner (Proposition 6.1).

Predictions to test, as in Remarks 4.6 and 5.3:

- the error floor rises once total samples fall below $\sim \kappa_{\text{coh}} d \log d$;

- preconditioning lowers the error at a fixed iteration count only while the leakage condition fails without it;
- the circulant limit is noisier than the $P = I$ limit;
- P_T reaches the $P = I$ limit in a few iterations.

]

8 Discussion and limitations

- **The denoiser hypothesis.** Assumption 4.1 (or Assumption 5.1) is a hypothesis about the trained denoiser. Approximate symmetry and projection-like spectra of denoiser Jacobians are well documented [32, 24, 31], but no theorem says that training produces small leakage. Leakage and Jacobian Lipschitzness can be measured with Jacobian-vector products, but only on samples, which gives lower bounds on suprema.
- **Global phase.** Theorem 5.2 covers the final, accuracy-determining phase. Convergence from a cold start (the analogue of support identification) is open. Theorem 5.5 reduces it to a deterministic statement about secants along long segments.
- **Momentum.** Accelerated variants break the one-step tangent structure (e_k is no longer the image of a single tangent map). A three-term version of Theorem 4.5 seems feasible. The degradation of accelerated PnP reported in [20] is a reason to do it.
- **Prior mismatch.** Mismatch enters only through ε_0 and μ. Distributional shift (phantom to patient) is not covered.
- **Hallucination.** The trade-off between passing the truth's tangent directions (ε_0 small) and suppressing null-space directions (η small) is a linearised form of the accuracy–stability trade-off of [16].

A Proof of Theorem 5.5

Step 1: one pair. Fix $T, T' \in \mathcal{T}$ and let $Y_j = \alpha \Pi_T B_j^* B_j \Pi_{T'}$, viewed as a $d \times d$ matrix from T' to T, and $Z_j = Y_j - \mathbb{E} Y_j$.

- **Uniform bound.** $\|Y_j\| \le \alpha \|B_j \Pi_T\| \|B_j \Pi_{T'}\| \le \alpha \Phi_{\mathcal{T}}$, so $\|Z_j\| \le 2\alpha \Phi_{\mathcal{T}}$.
- **Variance.** We have
$$\mathbb{E} Z_j Z_j^* \preceq \mathbb{E} Y_j Y_j^* \preceq \alpha^2 \Phi_{\mathcal{T}} \, \mathbb{E}\, \Pi_T B_j^* B_j \Pi_T,$$
so the variance proxy is at most $\alpha^2 N_b \Phi_{\mathcal{T}} \lambda_+(\mathcal{T})$. The same holds for $Z_j^* Z_j$.
- **Tail bound.** The rectangular matrix Bernstein inequality [42, Thm. 1.6] gives, for $\tau > 0$,
$$\Pr\Big[\big\|\textstyle\sum_j Z_j\big\| \ge \tau\Big] \le 2d \exp\Big(-\frac{\tau^2/2}{\alpha^2 N_b \Phi_{\mathcal{T}} \lambda_+ + 2\alpha \Phi_{\mathcal{T}} \tau/3}\Big) = 2d \exp\Big(-\frac{N_b \tau^2}{2a\Phi_{\mathcal{T}}(a\lambda_+ + 2\tau/3)}\Big).$$

Step 2: from a net to the family. Let $\Delta = \alpha(H - N_b \bar{H})$. On $\{\|\alpha H\| \le 2\}$, and since $a\lambda_{\max}(\bar{H}) \le 1$, we have $\|\Delta\| \le 3$. Let $\mathcal{N}_\epsilon \subset \mathcal{T}$ be an ϵ-net. For $T, T' \in \mathcal{T}$ with nearest net points $\tilde{T}, \tilde{T}'$,
$$\|\Pi_T \Delta \Pi_{T'}\| \le \|\Pi_{\tilde{T}} \Delta \Pi_{\tilde{T}'}\| + \|\Delta\|\big(\|\Pi_T - \Pi_{\tilde{T}}\| + \|\Pi_{T'} - \Pi_{\tilde{T}'}\|\big) \le \max_{\text{net}} + 6\epsilon.$$

Step 3: union bound. Take $\tau = t/2$ and $\epsilon = t/12$, and apply a union bound over the $|\mathcal{N}_\epsilon|^2$ pairs. Then $\sup_{T,T'} \|\Pi_T \Delta \Pi_{T'}\| \leq t$ except with probability

$$2d\mathcal{N}(\mathcal{T}, t/12)^2 \exp\big(-N_b t^2/(8a\Phi_{\mathcal{T}}(a\lambda_+ + t/3))\big) \leq p.$$

Finally, $\|\Pi_T(I - \alpha H)\Pi_{T'}\| \leq \|\Pi_T(I - a\bar{H})\Pi_{T'}\| + \|\Pi_T \Delta \Pi_{T'}\|$. □

References

[1] Ben Adcock, Juan M. Cardenas, and Nick Dexter. A unified framework for learning with nonlinear model classes from arbitrary linear samples, 2023. arXiv:2311.14886.

[2] Ben Adcock, Anders C. Hansen, Clarice Poon, and Bogdan Roman. Breaking the coherence barrier: a new theory for compressed sensing. *Forum of Mathematics, Sigma*, 5:e4, 2017.

[3] Richard G. Baraniuk and Michael B. Wakin. Random projections of smooth manifolds. *Foundations of Computational Mathematics*, 9:51–77, 2009.

[4] Jonathan Barzilai and Jonathan M. Borwein. Two-point step size gradient methods. *IMA Journal of Numerical Analysis*, 8(1):141–148, 1988.

[5] Stephen Becker and Jalal Fadili. A quasi-Newton proximal splitting method. In *Advances in Neural Information Processing Systems*, volume 25, 2012.

[6] Stephen Becker, Jalal Fadili, and Peter Ochs. On quasi-Newton forward-backward splitting: proximal calculus and convergence. *SIAM Journal on Optimization*, 29(4):2445–2481, 2019. CHECK pages.

[7] Aaron Berk, Simone Brugiapaglia, Babhru Joshi, Yaniv Plan, Matthew Scott, and Özgür Yılmaz. A coherence parameter characterizing generative compressed sensing with Fourier measurements. *IEEE Journal on Selected Areas in Information Theory*, 3(3):502–512, 2022. arXiv:2207.09340. CHECK pages.

[8] Thomas Blumensath and Mike E. Davies. Sampling theorems for signals from the union of finite-dimensional linear subspaces. *IEEE Transactions on Information Theory*, 55(4):1872–1882, 2009.

[9] Ashish Bora, Ajil Jalal, Eric Price, and Alexandros G. Dimakis. Compressed sensing using generative models. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *PMLR*, pages 537–546, 2017.

[10] Stéphane Boucheron, Gábor Lugosi, and Pascal Massart. *Concentration Inequalities: A Nonasymptotic Theory of Independence*. Oxford University Press, 2013.

[11] Kristian Bredies and Dirk A. Lorenz. Linear convergence of iterative soft-thresholding. *Journal of Fourier Analysis and Applications*, 14:813–837, 2008.

[12] Stanley H. Chan, Xiran Wang, and Omar A. Elgendy. Plug-and-play ADMM for image restoration: Fixed-point convergence and applications. *IEEE Transactions on Computational Imaging*, 3(1):84–98, 2017.

[13] William C. Davidon. Optimally conditioned optimization algorithms without line searches. *Mathematical Programming*, 9:1–30, 1975.

[14] Armin Eftekhari and Michael B. Wakin. New analysis of manifold embeddings and signal recovery from compressive measurements. *Applied and Computational Harmonic Analysis*, 39(1):67–109, 2015.

[15] Davis Gilton, Gregory Ongie, and Rebecca Willett. Deep equilibrium architectures for inverse problems in imaging. *IEEE Transactions on Computational Imaging*, 7:1123–1133, 2021.

[16] Nina M. Gottschling, Vegard Antun, Anders C. Hansen, and Ben Adcock. The troublesome kernel: on hallucinations, no free lunches, and the accuracy-stability tradeoff in inverse problems. *SIAM Review*, 2025. arXiv:2001.01258. CHECK volume and pages.

[17] Elaine T. Hale, Wotao Yin, and Yin Zhang. Fixed-point continuation for ℓ_1-minimization: methodology and convergence. *SIAM Journal on Optimization*, 19(3):1107–1130, 2008.

[18] Sébastien Herbreteau, Emmanuel Moebel, and Charles Kervrann. Normalization-equivariant neural networks with application to image denoising. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

[19] Tao Hong, Luis Hernandez-Garcia, and Jeffrey A. Fessler. A complex quasi-Newton proximal method for image reconstruction in compressed sensing MRI. *IEEE Transactions on Computational Imaging*, 10:372–384, 2024.

[20] Tao Hong, Xiaojian Xu, Jason Hu, and Jeffrey A. Fessler. Provable preconditioned plug-and-play approach for compressed sensing MRI reconstruction. *IEEE Transactions on Computational Imaging*, 10:1476–1488, 2024. arXiv:2405.03854.

[21] Samuel Hurault, Arthur Leclaire, and Nicolas Papadakis. Gradient step denoiser for convergent plug-and-play. In *International Conference on Learning Representations*, 2022.

[22] M. F. Hutchinson. A stochastic estimator of the trace of the influence matrix for Laplacian smoothing splines. *Communications in Statistics – Simulation and Computation*, 18(3):1059–1076, 1989.

[23] Joundi, Yann Traonmilin, and Jean-François Aujol. From sparse recovery to plug-and-play priors, understanding trade-offs for stable recovery with generalized projected gradient descent, 2025. arXiv:2512.07397. CHECK first author's given name.

[24] Zahra Kadkhodaie, Florentin Guth, Eero P. Simoncelli, and Stéphane Mallat. Generalization in diffusion models arises from geometry-adaptive harmonic representations. In *International Conference on Learning Representations*, 2024.

[25] Zahra Kadkhodaie and Eero P. Simoncelli. Solving linear inverse problems using the prior implicit in a denoiser. In *Advances in Neural Information Processing Systems*, volume 34, 2021. arXiv:2007.13640.

[26] Ulugbek S. Kamilov, Charles A. Bouman, Gregery T. Buzzard, and Brendt Wohlberg. Plug-and-play methods for integrating physical and learned models in computational imaging: Theory, algorithms, and applications. *IEEE Signal Processing Magazine*, 40(1):85–97, 2023.

[27] Felix Krahmer and Rachel Ward. Stable and robust sampling strategies for compressive imaging. *IEEE Transactions on Image Processing*, 23(2):612–622, 2014.

[28] Leong and Yann Traonmilin. A recovery theory for diffusion priors: deterministic analysis of the implicit prior algorithm, 2025. arXiv:2509.20511. CHECK first author's given name.

[29] Jingwei Liang, Jalal Fadili, and Gabriel Peyré. Activity identification and local linear convergence of forward–backward-type methods. *SIAM Journal on Optimization*, 27(1):408–437, 2017.

[30] Jiaming Liu, M. Salman Asif, Brendt Wohlberg, and Ulugbek S. Kamilov. Recovery analysis for plug-and-play priors using the restricted eigenvalue condition. In *Advances in Neural Information Processing Systems*, volume 34, 2021. arXiv:2106.03668.

[31] Hila Manor and Tomer Michaeli. On the posterior distribution in denoising: Application to uncertainty quantification. In *International Conference on Learning Representations*, 2024.

[32] Sreyas Mohan, Zahra Kadkhodaie, Eero P. Simoncelli, and Carlos Fernandez-Granda. Robust and interpretable blind image denoising via bias-free convolutional neural networks. In *International Conference on Learning Representations*, 2020.

[33] James M. Ortega and Werner C. Rheinboldt. *Iterative Solution of Nonlinear Equations in Several Variables*. Academic Press, 1970. Reprinted SIAM Classics in Applied Mathematics 30, 2000.

[34] M. R. Osborne and L. Sun. A new approach to symmetric rank-one updating. *IMA Journal of Numerical Analysis*, 19(4):497–507, 1999.

[35] Jean-Christophe Pesquet, Audrey Repetti, Matthieu Terris, and Yves Wiaux. Learning maximally monotone operators for image recovery. *SIAM Journal on Imaging Sciences*, 14(3):1206–1237, 2021.

[36] James G. Pipe and Padmanabhan Menon. Sampling density compensation in MRI: rationale and an iterative numerical solution. *Magnetic Resonance in Medicine*, 41(1):179–186, 1999.

[37] Klaas P. Pruessmann, Markus Weiger, Peter Börnert, and Peter Boesiger. Advances in sensitivity encoding with arbitrary k-space trajectories. *Magnetic Resonance in Medicine*, 46(4):638–651, 2001.

[38] Francesco Renna, Robert Calderbank, Lawrence Carin, and Miguel R. D. Rodrigues. Reconstruction of signals drawn from a Gaussian mixture via noisy compressive measurements. *IEEE Transactions on Signal Processing*, 62(9):2265–2277, 2014.

[39] Ernest Ryu, Jialin Liu, Sicheng Wang, Xiaohan Chen, Zhangyang Wang, and Wotao Yin. Plug-and-play methods provably converge with properly trained denoisers. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *PMLR*, pages 5546–5557, 2019.

[40] Charles M. Stein. Estimation of the mean of a multivariate normal distribution. *The Annals of Statistics*, 9(6):1135–1151, 1981.

[41] Yann Traonmilin, Jean-François Aujol, and Antoine Guennec. Towards optimal algorithms for the recovery of low-dimensional models with linear rates, 2024. arXiv:2410.06607.

[42] Joel A. Tropp. User-friendly tail bounds for sums of random matrices. *Foundations of Computational Mathematics*, 12:389–434, 2012.

[43] Singanallur V. Venkatakrishnan, Charles A. Bouman, and Brendt Wohlberg. Plug-and-play priors for model based reconstruction. In *IEEE Global Conference on Signal and Information Processing*, pages 945–948, 2013.

[44] Ke Wei, Jian-Feng Cai, Tony F. Chan, and Shingyu Leung. Guarantees of Riemannian optimization for low rank matrix recovery. *SIAM Journal on Matrix Analysis and Applications*, 37(3):1198–1222, 2016.

[45] Wolf, Jalal Fadili, Guo, and He. Physics matters in PnP: recovery guarantees with the MMSE and NN denoisers, 2026. arXiv:2607.29525. CHECK author given names.